25900522

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEMBG200 - BIOINSTRUMENTATION AND TECHNIQUES

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A - Very Short Answer **Answer all questions** **1X5=5 marks**

1. Why is a laser used as the light source in confocal microscopy? **[K] / [CO1]**

   For high intensity and to produce a pinpoint light source. **[1]**

2. Which cuvette is used in visible spectrophotometry **[K] / [CO4]**

   Glass or quartz cuvette. **[1]**

3. What is preparative centrifugation? [K] / [CO2]

   Used to separate and isolate specific biological particles based on size, shape & density **[1]**

4. Which type of repeat sequence is present in a VNTR? **[U] / [CO5]**

   Short tandem repeats (STRs) **[1]**

5. What is the purpose of the excitation filter in a fluorescence microscope? **[K] / [CO1]**

   To select the specific wavelength required to excite the fluorophore. **[1]**

Part B - Short Answer **Answer any 10 questions** **2X10 = 20 marks**

6. Explain how turbidity is related to microbial growth [U] / [CO4]

   Increased microbial cell density scatters more light, increasing the measured turbidity. **[2]**

7. Comment on Western blotting. [K] / [CO3]

   A technique for detecting specific proteins in a sample using antibody binding**. [2]**

8. Why can't live specimens be observed in SEM? [U] / [CO1]

Because the sample chamber must be under a high vacuum, which is incompatible with living, hydrated specimens. **[2]**

9. Why do smaller proteins migrate faster in PAGE? [U] / [CO3]

Smaller proteins move through the pores of the gel more easily than larger, bulkier proteins. **[2]**

10. Which biological molecules are separated using AGE? [K] / [CO3]

Nucleic acids (DNA and RNA). **[1+1]**

11. What is the role of immersion oil in microscopy, and why is it used? [k] / [CO1]

To increase resolution by preventing light refraction using a medium that has a similar refractive index to glass. **[2]**

12. State why temperature control is critical in Isoelectric Focusing [K] / [CO3]

To maintain a stable pH gradient, as heat generated by the electric current can disrupt the pH and ruin the separation. **[2]**

13. Summarise the steps of Southern blotting. [E] / [CO3]

1. Separate DNA on a gel. 2. Transfer to a membrane. 3. Probe with labelled DNA. 4. Detect the target sequence. **[2]**

14. What are VNTRs, and why are they important in DNA fingerprinting? [K] / [CO5]

Variable Number Tandem Repeats; they are highly polymorphic in length, creating unique individual patterns for identification. **[2]**

15. Explain the principle of centrifugation [U] / [CO2]

It uses centrifugal force to separate particles in a mixture [1] based on their size, density, and shape. **[1]**

16. List two advantages of light microscopy. [An]/ [CO1]

1. Ability to view living cells. 2. Natural colour visualisation**. [2]**

17. What type of samples might require nested PCR for detection? [A]/ [CO5]

Samples with very low amounts of the target DNA, such as in early pathogen detection or ancient DNA samples. **[2]**

Part C - Short Essay **Answer any 5 questions** **6X5 = 30 marks**

18. Explain the steps involved in Northern Blotting and its application in gene expression studies U] / [CO3]

- **Steps:** RNA extraction → Gel electrophoresis → Transfer to membrane → Hybridization with probe → Detection. **[4]**
- **Application:** Detects specific mRNA to study gene expression levels. **[2]**

19. Describe the optical components of a dark-field microscope and their role **[K]/ [CO1]**

Components: High-intensity light source, special condenser with opaque disk, and objective.**[3]**

- **Function:** Condenser blocks direct light; only scattered light enters the objective, making specimens bright against a dark background. **[3]**

20. Explain the causes and effects of chromatic aberration in lenses **[U]/ [CO1]**

- **Cause:** Lens refracts different wavelengths of light at various angles**.[3]**
- **Effect:** Colours focus at different points, causing blurred images with coloured fringes. **[3]**

21. Discuss the use of SDS-PAGE in Clinical Diagnostics **[A]/ [CO3]**

- **Use:** Separates proteins by molecular weight.**[3]**
- **Application:** Identifies disease biomarkers, analyses serum proteins, and detects abnormalities like monoclonal gammopathies. **[3]**

22. Describe the basic components of a UV spectrophotometer and how they work together to measure absorbance. **[U]/ [CO4]**

- **Components:** UV source, monochromator, cuvette, detector, readout.**[3]**
- **Workflow:** Light passes through the sample; detector measures transmitted light; absorbance calculated using Beer-Lambert law. **[3]**

23. Describe the working of a phase contrast microscope with the role of the annular diaphragm and phase plate. **[U]/ [CO1]**

- **Principle:** Converts phase differences in light into brightness contrast**.[2]**
- **Components:** Annular diaphragm creates hollow light cone; phase plate retards direct light, enhancing contrast via interference. **[4]**

24. Explain the principle and methodology of RFLP **[K]/ [CO5]**

- **Principle:** DNA variations alter restriction enzyme cut sites, changing fragment lengths.**[2]**
- **Methodology:** Digest DNA → Separate fragments by electrophoresis → Blot and hybridise with probe → Analyse band patterns. **[4]**

Part D - Long Essay Answer any 1 question 15X1 = 15 marks

25. Explain transmittance and absorbance—define both, derive their relationship, and interpret their significance in practical colourimetric analysis **[U]/ [CO4]**

- Definitions**:** Transmittance **(T):** Fraction of incident light that passes through a sample. $T = I/I_0$. **[2].** Absorbance **(A):** Logarithmic measure of light absorbed. $A = \log_{10}(I_0/I)$. **[2]**
- Relationship Derivation **:** $A = \log_{10}(I_0/I) = \log_{10}(1/T) = -\log_{10} T$. **[4]**
- Significance**:** Absorbance is directly proportional to concentration (Beer-Lambert Law: $A = \epsilon bc$). **[4].** This enables the use of a calibration curve to determine unknown concentrations, thereby forming the basis of quantitative analysis. **[3] -**

26. Describe the working principle of a Transmission Electron Microscope. Discuss its advantages and limitations **[U]/ [CO1]**

Working Principle **[5] &** Diagram **[2].** Advantages **[4**] Limitations **[4]**